

# Machine Learning — Lecture 02: Decision Trees and Bias-Variance Decomposition

المحاضرة الثانية: شجرة القرار (Decision Trees) وتحليل الـ Bias-Variance

## Part 1: Multiple Choice Questions (25)

### 1. In a decision tree, what do the internal nodes and the leaf nodes represent?

في الـ decision tree، إيه اللي بتمثله الـ internal nodes والـ leaf nodes؟

A) Internal nodes store predictions; leaf nodes test a feature

**B) Internal nodes test a feature; leaf nodes are the outputs (predictions)**

C) Internal nodes store the training labels; leaf nodes store hyperparameters

D) Both internal and leaf nodes only compute distances

الإجابة الصحيحة: B) Internal nodes test a feature; leaf nodes are the outputs (predictions)  
ليه صح: الـ internal node بتختبر feature والتفرّع بيعتمد على قيمته، أما الـ leaf node فهي المخرجات (النتيئات). العكس غلط، والمسافات من kNN مش من الشجرة.

Ref: Lecture 02 — Slide 7

### 2. How are continuous features split in a decision tree?

إزاي بيتم تقسيم الـ continuous features في الـ decision tree؟

A) By creating one branch for every distinct value of the feature

B) By computing the Euclidean distance to the nearest leaf

C) By removing the feature from the data before splitting

**D) By checking whether the feature is greater than or less than some threshold**

الإجابة الصحيحة: D) By checking whether the feature is greater than or less than some threshold  
ليه صح: الـ continuous feature بنقارنها بـ threshold (أكبر أو أصغر)، وبكده الـ decision boundary بتتكوّن من مستويات موازية للمحاور (axis-aligned). فرع لكل قيمة ده بيناسب الـ discrete features.

Ref: Lecture 02 — Slide 6

### 3. In a regression tree, how is the leaf value $y_m$ typically set?

في الـ regression tree، إزاي بتتحدد قيمة الـ leaf وهي  $y_m$  عادةً؟

A) To the most common target value in the region

**B) To the mean of the target values in the region**

C) To the largest target value in the region

D) To the number of examples in the region

الإجابة الصحيحة: B) To the mean of the target values in the region  
ليه صح: الـ regression tree بتديّ مخرجات مستمرة، فقيمة الـ leaf هي متوسط الـ targets في المنطقة. أما أكثر قيمة شيوعًا فهي للـ classification tree.

Ref: Lecture 02 — Slide 10

#### 4. What does each path from the root to a leaf define in a decision tree?

إليه الذي يحدده كل مسار من ال root لحد leaf في ال decision tree؟

##### A) A region $R_m$ of the input space

- B) A single training example
- C) A new hyperparameter
- D) A probability distribution over all features

الإجابة الصحيحة: A) A region  $R_m$  of the input space

ليه صح: كل مسار يحدد region  $R_m$  في فراغ المدخلات، والأمثلة التي تقع فيها هي التي بتحدد قيمة ال leaf.

Ref: Lecture 02 — Slide 10

#### 5. How are discrete features split in a decision tree?

إزاي بنقسم ال discrete features في ال decision tree؟

- A) By a single numeric threshold chosen on the feature
- B) By averaging the values of the feature over all examples
- C) Into a partition of the possible values of the feature
- D) By sorting the examples by their labels

الإجابة الصحيحة: C) Into a partition of the possible values of the feature

ليه صح: ال discrete feature بنقسم قيمها الممكنة إلى partition (مجموعات)، زي  $\text{Outlook} = \{\text{Sunny}, \text{Overcast}, \text{Rain}\}$ . ال threshold الرقمي لل continuous.

Ref: Lecture 02 — Slide 12

#### 6. What does the lecture say about constructing a tree with one leaf per training point?

إليه الذي بتقوله المحاضرة عن بناء شجرة فيها leaf لكل نقطة تدريب؟

- A) It is always possible, but it probably will not generalize
- B) It is impossible for most real datasets with many features
- C) It guarantees the best possible test performance
- D) It is only possible for regression problems

الإجابة الصحيحة: A) It is always possible, but it probably will not generalize

ليه صح: ال decision trees universal function approximators، فنقدر دايماً نبني شجرة بـ leaf لكل مثال، لكنها غالباً overfit ومنش هتعمم.

Ref: Lecture 02 — Slide 13

#### 7. What is the computational complexity class of finding the smallest decision tree that correctly classifies a training set?

ما هو تصنيف التعقيد الحسابي لإيجاد أصغر decision tree يصنف ال training set صح؟

- A) Solvable in linear time
- B) NP-complete
- C) Solvable in logarithmic time
- D) Solvable by one pass of gradient descent

الإجابة الصحيحة: B) NP-complete

ليه صح: إيجاد أصغر شجرة NP-complete، ولذلك بنستخدم greedy heuristic بدل البحث عن الحل الأمثل.

Ref: Lecture 02 — Slide 13

### 8. Which strategy is used to construct a useful decision tree in practice?

أي استراتيجية بنسخدمها لبناء decision tree مفيدة عملياً؟

**A) A greedy heuristic: pick the split that most reduces a loss, then recurse on the subpartitions**

- B) An exhaustive search over every possible tree for the given data
- C) A single split on a randomly chosen feature, with no recursion
- D) Averaging the labels of all training points to form one prediction

**الإجابة الصحيحة:** A) A greedy heuristic: pick the split that most reduces a loss, then recurse on the subpartitions  
**ليه صح:** بنسخدم greedy heuristic: نختار ال feature وال split اللي بيقلل ال loss أكثر، وبعدين نكرر على الأجزاء. البحث الشامل مكلف لأن المشكلة NP-complete.

Ref: Lecture 02 — Slide 14

### 9. Which leaf node has the highest uncertainty in its prediction?

أي leaf node يكون عدم اليقين في تنبؤه أعلى؟

- A) A leaf in which all examples have the same class
- B) A leaf in which 9 of the 10 examples are oranges
- C) A leaf in which each class has the same number of examples**
- D) A leaf whose entropy is exactly zero

**الإجابة الصحيحة:** C) A leaf in which each class has the same number of examples  
**ليه صح:** لو كل class ليه نفس عدد الأمثلة في ال leaf فعدم اليقين أعلى ما يمكن (high uncertainty). لو كل الأمثلة من نفس ال class فال uncertainty منخفضة ( $entropy = 0$ ).

Ref: Lecture 02 — Slide 18

### 10. How many bits of entropy does a fair coin flip have?

كام bit من ال entropy في رمية عملة عادلة؟

- A) 1 bit**
- B) 0 bits
- C) 2 bits
- D) 0.5 bits

**الإجابة الصحيحة:** A) 1 bit

**ليه صح:** ال fair coin ( $p = 0.5$ ) أقصى عدم يقين ممكن لمتغير ثنائي، فال  $entropy = -0.5 \log_2(0.5) - 0.5 \log_2(0.5) = 1 \text{ bit}$ . ال entropy بتبقى 0 فقط لما  $p = 0$  أو 1.

Ref: Lecture 02 — Slide 22

### 11. For regression trees, how are the splits chosen?

في ال regression trees، إزاي بيتم اختيار ال splits؟

- A) To minimize squared error rather than maximize information gain**
- B) To maximize information gain, exactly as in classification
- C) To maximize the number of leaves
- D) To make all leaf values equal

**الإجابة الصحيحة:** A) To minimize squared error rather than maximize information gain  
**ليه صح:** ال regression trees بتختار ال splits اللي بتقلل ال squared error بدل ما تعظم ال information gain (اللي يُستخدم في ال classification).

Ref: Lecture 02 — Slide 40

## 12. What did Claude Shannon show about storing the outcome of a random draw?

إيه اللي أثبتته Claude Shannon عن تخزين نتيجة سحب عشوائي؟

- A) Any random outcome can always be stored using exactly one bit of memory
- B) The entropy of a variable is always equal to its number of outcomes
- C) Storing outcomes always requires more bits than the number of outcomes

**D) You cannot store it using fewer expected bits than the entropy without losing information**

الإجابة الصحيحة: D) You cannot store it using fewer expected bits than the entropy without losing information  
ليه صح: Shannon أثبت إن مينفعش نخزن النتيجة بعدد bits متوقع أقل من ال entropy من غير فقدان معلومات، عشان كده وحدة ال entropy هي bits.

Ref: Lecture 02 — Slide 22

## 13. What does the information gain of an attribute measure?

إيه اللي بقيسه ال information gain ل attribute؟

- A) The total number of leaves produced by the split in the tree

**B) The expected reduction in entropy caused by partitioning on that attribute**

- C) The Euclidean distance between the two partitions of the data
- D) The accuracy of the final tree on the held-out test set

الإجابة الصحيحة: B) The expected reduction in entropy caused by partitioning on that attribute  
ليه صح: ال information gain هو التخفيض المتوقع في ال entropy (عدم اليقين عن ال label) بعد التقسيم على ال attribute، وهو بالطبط اللي نحتاجه لاختيار أفضل split.

Ref: Lecture 02 — Slide 31

## 14. In $\text{Gain}(S, a) = \text{Entropy}(S) - \sum_v \frac{|S_v|}{|S|} \text{Entropy}(S_v)$ , how is the entropy of the partitions combined?

في قانون  $\text{Gain}(S, a)$ ، إزاي بيتم دمج entropy الأجزاء؟

- A) The entropies of the partitions are averaged without any weights
- B) Only the entropy of the largest partition is used in the formula

**C) Each partition's entropy is weighted by its size relative to the original set**

- D) The entropies of the partitions are simply summed over all values of  $v$

الإجابة الصحيحة: C) Each partition's entropy is weighted by its size relative to the original set  
ليه صح: كل جزء بيتوزن بحجمه نسبةً للمجموعة الأصلية  $|S_v| / |S|$ ، فالجزء الأكبر بيأخذ وزن أكبر. المتوسط غير الموزون أو المجموع بيدي نتيجة غلط.

Ref: Lecture 02 — Slide 31

## 15. In the Play Tennis example, which feature has the highest information gain and is chosen for the root split?

في مثال Play Tennis، أي feature ليه أعلى information gain ويختار لل root؟

- A) Humidity, with a gain of about 0.151

**B) Outlook, with a gain of about 0.246**

- C) Wind, with a gain of about 0.048
- D) Temperature, with a gain of about 0.029

الإجابة الصحيحة: B) Outlook, with a gain of about 0.246  
ليه صح: ال gains: Outlook 0.246 و Humidity 0.151 و Wind 0.048 و Temperature 0.029، ف Outlook هو الأعلى فيختار لل split الأول.

Ref: Lecture 02 — Tennis example slides (Play Tennis, Information Gain)

**16. In the Play Tennis data (9 positive and 5 negative examples), what is the entropy of the target Play?**

في بيانات Play Tennis (9 أمثلة موجبة و5 سالبة)، كام entropy المتغير Play؟

**A) About 0.94**

B) About 0.50

C) Exactly 1.00

D) Exactly 0

الإجابة الصحيحة: A) About 0.94

ليه صح:  $p+ = 9/14$  و  $p- = 5/14$  فال  $\text{entropy} = -(9/14)\log_2(9/14) - (5/14)\log_2(5/14) \approx 0.94$ . هي مش 1 لأن التوزيع مش متساوي ومش 0 لأن فيه classes مختلطة.

Ref: Lecture 02 — Tennis example slides (Play Tennis, Information Gain)

**17. In the Play Tennis tree, why does the Overcast branch of Outlook become a leaf that predicts Yes?**

في شجرة Play Tennis، ليه فرع Overcast بقى leaf بيتنبأ بـ Yes؟

A) Overcast has the lowest number of attributes

B) Overcast has equal numbers of positive and negative examples

C) The Overcast branch contains no examples

**D) All four Overcast examples are positive, so its entropy is 0**

الإجابة الصحيحة: D) All four Overcast examples are positive, so its entropy is 0

ليه صح: كل أمثلة Overcast الأربعة موجبة (+, -0) فال  $\text{entropy} = 0$  ولا يوجد داعي للتقسيم. لو كانت متساوية كانت ال entropy أعلى ما يمكن.

Ref: Lecture 02 — Tennis example slides (Play Tennis, Information Gain)

**18. In the decision tree construction algorithm, what is returned for a group of examples that is empty?**

في خوارزمية بناء ال decision tree، إيه اللي بيتم إرجاعه لمجموعة أمثلة فاضية؟

**A) The majority class from the parent**

B) The class with the fewest examples overall

C) A random class label

D) The feature with the highest gain

الإجابة الصحيحة: A) The majority class from the parent

ليه صح: لو مفيش أمثلة في المجموعة بنرجع ال majority class من ال parent. لو كل الأمثلة من نفس ال class بنرجع ال class ده، ولو لا بنكمل التقسيم.

Ref: Lecture 02 — Slide 35

**19. Which statement about conditional entropy is correct?**

أي عبارة عن ال conditional entropy صحيحة؟

A) Knowing X can increase the uncertainty about Y:  $H(Y|X) > H(Y)$

B)  $H(Y|Y) = 1$  for every non-constant random variable Y

C) If X and Y are independent, then  $H(Y|X) = 0$  exactly

**D) By knowing X we can only decrease our uncertainty about Y:  $H(Y|X) \leq H(Y)$**

الإجابة الصحيحة: D) By knowing X we can only decrease our uncertainty about Y:  $H(Y|X) \leq H(Y)$

ليه صح: معرفة X ممكن تقلل عدم اليقين عن Y بس مش تزوده، ف  $H(Y|X) \leq H(Y)$  و  $H(Y|Y) = 0$  مش 1، ولو X و Y مستقلين ف  $H(Y|X) = H(Y)$ .

Ref: Lecture 02 — Slide 29

## 20. In the restaurant example, what are the information gains of the attributes Type and Patrons?

في مثال المطعم، كام ال information gain لل attributes: Type و Patrons؟

- A)  $IG(\text{Type})$  is about 0.541 and  $IG(\text{Patrons}) = 0$
- B) Both  $IG(\text{Type})$  and  $IG(\text{Patrons})$  are equal to 1
- C)  $IG(\text{Type}) = 0$  and  $IG(\text{Patrons})$  is about 0.541**
- D) Both  $IG(\text{Type})$  and  $IG(\text{Patrons})$  are equal to 0

الإجابة الصحيحة: C)  $IG(\text{Type}) = 0$  and  $IG(\text{Patrons})$  is about 0.541

ليه صح: Type ملهوش أي قدرة تميز ( $IG = 0$ ) لأن كل نوع فيه نص أمثلة موجبة ونصها سالبة، بينما  $\text{Patrons} \approx 0.541$  فهو الأفضل. الاختيارات الثانية بتعكس أو بتساوي القيم غلط.

Ref: Lecture 02 — Slide 37

## 21. According to Occam's razor, which hypothesis should we prefer?

حسب Occam's razor، أي فرضية نفضلها؟

- A) The most complex hypothesis that fits the training data
- B) The hypothesis with the largest number of parameters
- C) The hypothesis that ignores the observations
- D) The simplest hypothesis that fits the observations**

الإجابة الصحيحة: D) The simplest hypothesis that fits the observations

ليه صح: المبدأ يقول نبحت عن أبسط فرضية تفسر الملاحظات. لهذا نفضل trees صغيرة بنودها المعلوماتية قريبة من ال root، مع صعوبة تعريف 'البساطة' رسميًا.

Ref: Lecture 02 — Slide 39

## 22. Which of the following is an advantage of decision trees over kNN?

أي من التالي ميزة لل decision trees على kNN؟

- A) They have very few hyperparameters
- B) They can easily incorporate interesting distance measures such as shape contexts
- C) They are fast at test time and more interpretable**
- D) They can never overfit the training data

الإجابة الصحيحة: C) They are fast at test time and more interpretable

ليه صح: الشجرة سريعة وقت الاختبار وأسهل في التفسير. قلة ال hyperparameters وإمكانية استخدام مقاييس مسافة خاصة هما مميزات kNN، والشجرة ممكن تعمل overfit.

Ref: Lecture 02 — Slide 41

## 23. For an ensemble of classifiers to be nontrivial, the classifiers must:

عشان ال ensemble يبقى مفيد (nontrivial)، لازم المصنفات:

- A) be exactly identical copies of one classifier trained on the same data
- B) always use the same training data and the same hyperparameters
- C) differ somehow, for example in algorithm, hyperparameters, data or weighting**
- D) be limited to decision trees of depth one trained on the same data

الإجابة الصحيحة: C) differ somehow, for example in algorithm, hyperparameters, data or weighting

ليه صح: لو المصنفات متطابقة فالتصويت بينها مفيهوش جديد. لازم تختلف في الخوارزمية أو ال hyperparameters أو بيانات التدريب أو أوزان الأمثلة.

Ref: Lecture 02 — Slide 42

24. In the decomposition  $E[(y - t)^2] = (y^* - E[y])^2 + \text{Var}(y) + \text{Var}(t)$ , what are the three terms called, in order?

في تفكيك الـ expected loss، إيه أسماء الحدود الثلاثة بالترتيب؟

- A) Variance, bias, and training error
- B) Bias, variance, and Bayes error**
- C) Underfitting, overfitting, and test error
- D) Mean, median, and mode

الإجابة الصحيحة: B) Bias, variance, and Bayes error

ليه صح: الحد الأول bias (مدى خطأ التنبؤ المتوقع، يقابل underfitting)، والثاني variance (تذبذب التنبؤات، يقابل overfitting)، والثالث Bayes error (عدم القابلية الذاتية للتنبؤ بالـ targets).

Ref: Lecture 02 — Slide 54

25. Under squared error loss, which prediction  $y^*$  is the best possible, given the conditional distribution  $p(t|x)$ ?

تحت squared error loss، أي تنبؤ  $y^*$  هو الأفضل إذا عرفنا التوزيع الشرطي  $p(t|x)$ ؟

- A)  $y^* = \text{Var}[t | x]$
- B)  $y^* = \text{the largest value of } t$
- C)  $y^* = 0 \text{ for every } x$
- D)  $y^* = E[t | x]$**

الإجابة الصحيحة: D)  $y^* = E[t | x]$

ليه صح: الإثبات بيوضح إن  $E[(y - t)^2 | x] = (y - y^*)^2 + \text{Var}[t | x]$  وبالتالي أفضل  $y$  هو  $y^* = E[t | x]$ . الحد الثاني  $\text{Var}[t|x]$  هو الـ Bayes error ومبعتمدش على  $y$ .

Ref: Lecture 02 — Slide 51

## Part 2: Essay and Calculation Questions (15)

1. Describe the structure of a decision tree and explain how it makes predictions for continuous and discrete features, including the difference between regression and classification trees.

صف بنية الـ decision tree وشرح إزاي بيعمل تنبؤات مع الـ continuous والـ discrete features، وإيه الفرق بين regression وclassification trees.

**Model answer:**

A decision tree makes predictions by splitting on features according to a tree structure. Internal nodes test a feature, branching is determined by the feature value, and leaf nodes are the outputs (predictions).

Continuous features: split by checking whether the feature is greater or less than a threshold; the decision boundary is made of axis-aligned planes.

Discrete features: split into a partition of the possible values.

Each path from the root to a leaf defines a region  $R_m$  of the input space containing training examples.

Regression tree: continuous output; the leaf value  $y_m$  is typically the mean of the targets in  $R_m$ .

Classification tree: discrete output; the leaf value is typically the most common label in  $R_m$ .

Ref: Lecture 02 — Slides 4-12

## 2. Why do we use a greedy heuristic to learn decision trees? Describe the greedy procedure.

ليه بنستخدم greedy heuristic لتعلم ال decision trees؟ اشرح خطوات الإجراء.

### Model answer:

Decision trees are universal function approximators: for any training set we can build a tree with exactly one leaf per training point, but it probably will not generalize. Finding the smallest decision tree that correctly classifies a training set is NP-complete.

So we resort to a greedy heuristic:

1. Start with the whole training set and an empty tree.
2. Pick a feature and a candidate split that would most reduce a loss (a scalar number: low is good, high is bad).
3. Split on that feature and recurse on the sub-partitions.

Greedy algorithms do not necessarily yield the global optimum.

Ref: Lecture 02 — Slides 13-14 and 40

## 3. Define entropy. Use the two-coin example to explain what high and low entropy mean and state the units of entropy.

عرّف ال entropy. استخدم مثال العمليتين لتوضيح معنى ال entropy العالية والمنخفضة واذكر وحدة القياس.

### Model answer:

Entropy is a property of a random variable: a number that quantifies the uncertainty inherent in its possible outcomes. For a discrete variable  $Y$ ,  $H(Y) = -\sum_y p(y) \log_2 p(y)$ . For a loaded coin with probability  $p$  of heads:  $-p \log_2(p) - (1-p) \log_2(1-p)$ .

The coin whose outcomes are more certain has lower entropy; for  $p = 0$  or  $p = 1$  the entropy is 0. A fair coin has 1 bit.

High entropy: uniform-like distribution over many outcomes, flat histogram, less predictable samples.

Low entropy: distribution concentrated on a few outcomes, more predictable samples.

Units: bits (Shannon: you cannot store a random outcome with fewer expected bits than the entropy without losing information). Entropy can also be seen as the expected information content of a random draw.

Ref: Lecture 02 — Slides 19-23

## 4. Explain information gain and why it is used to choose splits in a decision tree. Write the formula and the main properties of conditional entropy used with it.

اشرح ال information gain وليه يُستخدم لاختيار ال splits في ال decision tree. اكتب القانون وأهم خصائص ال conditional entropy.

### Model answer:

Information gain measures how much information about the class label  $Y$  is gained by knowing which side of a split we are on, i.e., the expected reduction in entropy:  $IG(Y|X) = H(Y) - H(Y|X)$  (the mutual information of  $Y$  and  $X$ ).

For an attribute  $a$ :  $Gain(S, a) = Entropy(S) - \sum_v \left( \frac{|S_v|}{|S|} \right) Entropy(S_v)$ , weighting each partition by its relative size.

If  $X$  is completely uninformative about  $Y$ :  $IG = 0$ . If  $X$  is completely informative:  $IG = H(Y)$ .

Properties:  $H$  is non-negative; chain rule  $H(X,Y) = H(X|Y) + H(Y) = H(Y|X) + H(X)$ ; if  $X$  and  $Y$  are independent  $H(Y|X) = H(Y)$ ;  $H(Y|Y) = 0$ ;  $H(Y|X) \leq H(Y)$ .

At each node we choose the feature (and threshold) with the highest gain.

Ref: Lecture 02 — Slides 24-31 and 34

**5. Explain the Play Tennis example: compute the root entropy, the gain of each feature, the root choice and the resulting tree.**

اشرح مثال Play Tennis: ال entropy لل root، وال gain لكل feature، واختيار ال root، والشجرة الناتجة.

**Model answer:**

The data has 14 examples (9 positive, 5 negative), so  $\text{Entropy}(\text{Play}) = 0.94$ .

Information gains: Outlook 0.246, Humidity 0.151, Wind 0.048, Temperature 0.029, so we split on Outlook.

Outlook branches: Sunny (2+, 3-), Overcast (4+, 0-), Rain (3+, 2-). Overcast is pure, so it becomes a leaf: Yes.

Sunny subset:  $\text{Gain}(\text{Humidity}) = 0.97$ ,  $\text{Gain}(\text{Temp}) = 0.57$ ,  $\text{Gain}(\text{Wind}) = 0.02$ , so split on Humidity: High -> No, Normal -> Yes.

Rain subset: split on Wind: Strong -> No, Weak -> Yes.

Final tree: Outlook at the root, Humidity under Sunny, Wind under Rain.

Ref: Lecture 02 — Tennis example slides (Play Tennis, Information Gain)

**6. Write and explain the decision tree construction algorithm (induceDecisionTree) and its stopping conditions.**

اكتب واشرح خوارزمية بناء ال decision tree (induceDecisionTree) وشروط التوقف.

**Model answer:**

Simple, greedy, recursive approach that builds the tree node by node:

1. If all examples in  $S$  have the same label  $y$ , return a leaf with that label.
2. Otherwise find the feature with the most information gain:  $i = \text{argmax}_i \text{Gain}(S, X_i)$ .
3. Split the examples into groups by the value of that feature.
4. For each group: if it has no examples, return the majority class from the parent; else if all examples are in the same class, return the class; otherwise recurse (induceDecisionTree on the subset).

Termination: when all leaves contain only examples of the same class or are empty (or every attribute is already included in the path).

Questions: how to choose the feature to split on, and how to choose the threshold for continuous features (the threshold that maximizes information gain).

Ref: Lecture 02 — Slides 35, 41 and 40

**7. What makes a good decision tree? Discuss Occam's razor and the main problems of decision trees.**

إيه اللي بيخلي ال decision tree كويسة؟ ناقش Occam's razor والمشاكل الأساسية لل decision trees.

**Model answer:**

A good tree is not too small: it must handle important but subtle distinctions in the data. It is also not too big: for computational efficiency (avoid redundant or spurious attributes), to avoid over-fitting the training examples, and for human interpretability.

Occam's razor: find the simplest hypothesis that fits the observations. It is useful but hard to formalize (how to define simplicity?).

We desire small trees with informative nodes near the root.

Problems: exponentially less data at lower levels; too big a tree can overfit; greedy algorithms do not necessarily yield the global optimum.

Ref: Lecture 02 — Slides 39-40

## 8. Compare decision trees with kNN: state the advantages of each.

قارن بين ال decision trees وkNN: اذكر مميزات كل واحد.

### Model answer:

Advantages of decision trees over kNN: simple to deal with discrete features, missing values, and poorly scaled data; fast at test time; more interpretable.

Advantages of kNN over decision trees: few hyperparameters; can incorporate interesting distance measures (e.g., shape contexts).

Ref: Lecture 02 — Slide 41

## 9. What is an ensemble of classifiers? Why must its members differ, and in what ways can they differ?

إيه هو ال ensemble من المصنفات؟ ولماذا لازم تختلف أعضاؤه وبأي طرق ممكن تختلف؟

### Model answer:

An ensemble is a set of predictors whose individual decisions are combined in some way to classify new examples, e.g., a (possibly weighted) majority vote.

For this to be nontrivial, the classifiers must differ somehow, for example: a different algorithm, a different choice of hyperparameters, training on different data, or training with a different weighting of the training examples.

Specific ensembling techniques are studied in the next lecture, and the bias-variance decomposition helps us understand them.

Ref: Lecture 02 — Slides 42 and 44

## 10. Explain the bias-variance decomposition of the expected squared error, including the proof that $y^* = E[t|x]$ is the best prediction and the meaning of each term.

اشرح تفكيك ال bias-variance لل expected squared error، مع إثبات إن  $y^* = E[t|x]$  هو أفضل تنبؤ ومعنى كل حد.

### Model answer:

Setup: fix a query point  $x$ ; repeatedly sample a training set  $D$  i.i.d. from  $p_{\text{sample}}$ , run the learner to get a prediction  $y$  at  $x$  (a random variable, since the randomness comes from the dataset), sample the true target  $t$  from  $p(t|x)$ , and compute the loss.  $y$  is independent of  $t$ .

Optimal prediction for squared error:  $E[(y - t)^2 | x] = y^2 - 2y E[t|x] + E[t^2|x] = (y - y^*)^2 + \text{Var}[t|x]$ , with  $y^* = E[t|x]$ . The first term is zero when  $y = y^*$ ; the second term is the Bayes error (noise, inherent unpredictability of  $t$ ), independent of  $y$ ; it is the best any algorithm can do. An algorithm achieving it is Bayes optimal.

Treating  $y$  as random:  $E[(y - t)^2] = (y^* - E[y])^2 + \text{Var}(y) + \text{Var}(t)$ .

Bias =  $(y^* - E[y])^2$ : how wrong the expected prediction is (underfitting). Variance =  $\text{Var}(y)$ : variability of predictions across datasets (overfitting). Bayes error =  $\text{Var}(t)$ : inherent unpredictability of the targets.

Overly simple models underfit and overly complex models overfit.

Ref: Lecture 02 — Slides 44-55

**11. Calculation: Compute the entropy (in bits) of three leaf nodes: A with 3 positive and 1 negative examples, B with 2 positive and 2 negative examples, and C with 4 positive and 0 negative examples.**

مسألة حسابية: احسب الـ entropy (بالـ bits) لثلاث leaf nodes: A فيها 3 موجب و1 سالب، B فيها 2 موجب و2 سالب، وC فيها 4 موجب و0 سالب.

**Model answer:**

Formula:  $H = -p+ \log_2(p+) - p- \log_2(p-)$ , with  $0 \log_2 0 = 0$ .

A:  $p+ = 3/4$ ,  $p- = 1/4$ :  $H = -0.75 \log_2(0.75) - 0.25 \log_2(0.25) = 0.311 + 0.5 = 0.811$  bits.

B:  $p+ = p- = 1/2$ :  $H = 0.5 + 0.5 = 1$  bit (maximum uncertainty).

C:  $p+ = 1$ ,  $p- = 0$ :  $H = 0$  bits (pure leaf, no uncertainty).

So B is the most uncertain leaf and C the most certain.

Ref: Lecture 02 — Slides 18-23

**12. Calculation: In the Play Tennis data (entropy 0.940), compute the information gain of Wind. Wind = Weak has 8 examples (6 positive, 2 negative) and Wind = Strong has 6 examples (3 positive, 3 negative).**

مسألة حسابية: في بيانات Play Tennis (الـ entropy = 0.940)، احسب الـ information gain لـ Wind. Wind = Weak فيها 8 أمثلة (6 موجب و2 سالب) و Wind = Strong فيها 6 أمثلة (3 موجب و3 سالب).

**Model answer:**

$\text{Entropy(Weak)} = -(6/8) \log_2(6/8) - (2/8) \log_2(2/8) = 0.811$ .

$\text{Entropy(Strong)} = -(3/6) \log_2(3/6) - (3/6) \log_2(3/6) = 1.0$ .

$\text{Expected entropy} = (8/14) \times 0.811 + (6/14) \times 1.0 = 0.463 + 0.429 = 0.892$ .

$\text{Gain(S, Wind)} = 0.940 - 0.892 = 0.048$ .

This matches the lecture (Wind: 0.048), which is much lower than Outlook (0.246).

Ref: Lecture 02 — Tennis example slides (Play Tennis, Information Gain)

**13. Calculation: In the Play Tennis data (entropy 0.940), compute the information gain of Temperature. Hot has 4 examples (2+, 2-), Mild has 6 examples (4+, 2-), and Cool has 4 examples (3+, 1-).**

مسألة حسابية: في بيانات Play Tennis (الـ entropy = 0.940)، احسب الـ information gain لـ Temperature. Hot فيها 4 أمثلة (2+ و2-)، Mild فيها 6 أمثلة (4+ و2-)، وCool فيها 4 أمثلة (3+ و1-).

**Model answer:**

$\text{Entropy(Hot)} = 1.0$  (2+, 2-).

$\text{Entropy(Mild)} = -(4/6) \log_2(4/6) - (2/6) \log_2(2/6) = 0.918$ .

$\text{Entropy(Cool)} = -(3/4) \log_2(3/4) - (1/4) \log_2(1/4) = 0.811$ .

$\text{Expected entropy} = (4/14) \times 1.0 + (6/14) \times 0.918 + (4/14) \times 0.811 = 0.286 + 0.393 + 0.232 = 0.911$ .

$\text{Gain(S, Temperature)} = 0.940 - 0.911 = 0.029$ .

This matches the lecture and is the lowest gain among the four features.

Ref: Lecture 02 — Tennis example slides (Play Tennis, Information Gain)

**14. Calculation: For the joint distribution of Raining (X) and Cloudy (Y):  $P(\text{Raining, Cloudy}) = 24/100$ ,  $P(\text{Raining, Not cloudy}) = 1/100$ ,  $P(\text{Not raining, Cloudy}) = 25/100$ ,  $P(\text{Not raining, Not cloudy}) = 50/100$ . Compute  $H(Y|X = \text{raining})$ ,  $H(Y|X)$  and the information gain  $IG(Y|X)$ .**

مسألة حسابية: للتوزيع المشترك بين (X) Raining و (Y) Cloudy المذكور بالأرقام أعلاه، احسب  $H(Y|X = \text{raining})$  و  $H(Y|X)$  و  $IG(Y|X)$ .

**Model answer:**

$$p(x = \text{raining}) = 25/100 = 1/4; p(x = \text{not raining}) = 75/100 = 3/4.$$

$$H(Y|X = \text{raining}) = -(24/25) \log_2(24/25) - (1/25) \log_2(1/25) = 0.24 \text{ bits.}$$

$$H(Y|X = \text{not raining}) = -(25/75) \log_2(25/75) - (50/75) \log_2(50/75) = 0.918 \text{ bits.}$$

$$H(Y|X) = (1/4)(0.24) + (3/4)(0.918) \approx 0.75 \text{ bits.}$$

$$H(Y): P(\text{cloudy}) = 49/100, \text{ so } H(Y) = H(0.49) \approx 1.0 \text{ bit.}$$

$$IG(Y|X) = H(Y) - H(Y|X) \approx 1.0 - 0.75 = 0.25 \text{ bits.}$$

$$\text{Check with the chain rule: } H(X, Y) = H(Y|X) + H(X) = 0.75 + 0.811 \approx 1.56 \text{ bits (matches the lecture).}$$

Ref: Lecture 02 — Slides 25-30

**15. Calculation: At a query point  $x$  the Bayes-optimal prediction is  $y^* = 5$ , the expected prediction of the learner is  $E[y] = 3$ , its variance is  $\text{Var}(y) = 0.5$ , and the target noise is  $\text{Var}(t) = 1$ . Compute the expected squared loss and each of its terms. Also find the leaf value of a regression tree whose region contains the targets  $\{2, 4, 6\}$ .**

مسألة حسابية: عند نقطة استعلام  $x$ ، التنبؤ الأمثل  $y^* = 5$ ، والتنبؤ المتوقع للتعلم  $E[y] = 3$ ، والتباين  $\text{Var}(y) = 0.5$ ، وضوضاء الهدف  $\text{Var}(t) = 1$ . احسب الخسارة المربعة المتوقعة وكمياتها، وكمية قيمة leaf لـ regression tree لمنطقتها فيها الـ targets  $\{2, 4, 6\}$ .

**Model answer:**

$$\text{Formula: } E[(y - t)^2] = (y^* - E[y])^2 + \text{Var}(y) + \text{Var}(t).$$

$$\text{Bias term} = (5 - 3)^2 = 4.$$

$$\text{Variance term} = 0.5.$$

$$\text{Bayes error} = 1 \text{ (cannot be reduced by any learner).}$$

$$\text{Expected loss} = 4 + 0.5 + 1 = 5.5.$$

$$\text{If the learner had } E[y] = y^*, \text{ the bias would be 0 and the loss would be } 0.5 + 1 = 1.5.$$

$$\text{Regression tree leaf value} = \text{mean of the targets} = (2 + 4 + 6) / 3 = 4.$$

Ref: Lecture 02 — Slides 10 and 53-54